

Florida Hurricane Catastrophe Loss Model

Supervisor Brief: Proposed Model Choice and Logic

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1 Project objective

The aim is not to replicate a vendor catastrophe model. The aim is to build an interpretable, reproducible hurricane loss model that explains the core catastrophe modelling chain:

event $\rightarrow$ hazard at location $\rightarrow$ vulnerability $\rightarrow$ insured loss $\rightarrow$ portfolio risk metrics.

Available data:

- **Claims:** hurr_id, property_id, loss.
- **Exposure:** property location, construction, occupancy, year built, replacement value, deductible, stories, gross area.
- **Hurricane track data:** storm snapshots with latitude/longitude, wind, pressure, category, radius of maximum winds, storm speed/direction, and quadrant wind radii.

The model should estimate event-level and portfolio-level losses for historical or hypothetical storms, with uncertainty and clear limitations.

2 Recommended model: two-part frequency-severity loss model

Use a **two-part model**: first estimate whether a property makes a claim, then estimate the size of the loss conditional on a claim. This is preferable to modelling loss ratio directly because the exposure data contains many property-storm pairs with zero loss. A single severity-only regression mixes two different questions:

Did the property suffer an insured loss? and How large was the loss if it did?

For property i and hurricane event j , define:

$$\begin{aligned} C_{ij} &= \mathbf{1}\{\text{property } i \text{ has a positive claim under hurricane } j\}, \\ L_{ij} &= \text{observed claim loss}, \\ V_i &= \text{total replacement value}, \\ D_i &= \text{deductible}, \\ Y_{ij} &= \frac{L_{ij}}{V_i} \quad \text{loss ratio, conditional on } C_{ij} = 1. \end{aligned}$$

The model is:

$$\mathbb{E}[L_{ij}^{\text{ground}}] = \Pr(C_{ij} = 1) \times \mathbb{E}[Y_{ij} \mid C_{ij} = 1] \times V_i.$$

This is exactly the intuition of a frequency-severity insurance model: claim probability times conditional severity times insured value.

3 Frequency model: claim occurrence

The frequency component estimates the probability that property i makes a claim under storm j :

$$p_{ij} = \Pr(C_{ij} = 1).$$

Use a logistic regression or regularised logistic regression:

$$\text{logit}(p_{ij}) = \alpha_0 + f_H(H_{ij}) + \alpha^\top X_i.$$

Definitions:

- H_{ij} : hazard features for property i under storm j .
- X_i : exposure/vulnerability features of property i .
- $f_H(\cdot)$: simple nonlinear hazard transform, e.g. splines, bins, or interactions. Keep it explainable.

Candidate hazard features:

- minimum distance from property to storm track;
- maximum nearby recorded storm wind;
- minimum central pressure, maximum SSHS category, radius of maximum winds;
- storm side/quadrant relative to track direction;
- whether property lies inside approximate 34/50/64 knot wind radii;
- distance to landfall or distance to coast if available/derivable.

Candidate exposure features:

- construction, occupancy, year built or age, stories, gross area;
- $\log(V_i)$, deductible ratio D_i/V_i ;
- location variables such as county/area, if not too granular.

Interpretation: the model estimates how storm intensity and proximity affect whether a claim occurs, adjusted for building vulnerability.

4 Severity model: loss ratio conditional on claim

For observations with $C_{ij} = 1$, model the conditional loss ratio:

$$Y_{ij} = \frac{L_{ij}}{V_i}, \quad Y_{ij} > 0.$$

A simple defensible specification is a Gamma GLM with log link:

$$\mathbb{E}[Y_{ij} \mid C_{ij} = 1] = \mu_{ij}, \quad \log(\mu_{ij}) = \beta_0 + g_H(H_{ij}) + \beta^\top X_i.$$

Why Gamma/log link:

- losses are positive and right-skewed;
- predictions remain positive;
- coefficients are interpretable as multiplicative effects;
- simpler to explain than black-box models.

Alternative: lognormal regression on $\log(Y_{ij})$. This is also acceptable, but Gamma GLM directly models the mean loss ratio, which is useful for pricing and portfolio loss estimation.

5 Financial loss calculation

The basic ground-up expected loss is:

$$\hat{L}_{ij}^{\text{ground}} = \hat{p}_{ij} \hat{\mu}_{ij} V_i.$$

A simple deductible-adjusted approximation is:

$$\hat{L}_{ij}^{\text{insured}} = \hat{p}_{ij} \max\{0, \hat{\mu}_{ij} V_i - D_i\}.$$

A better version for the final model is simulation-based:

1. draw $C_{ij}^{(s)} \sim \text{Bernoulli}(\hat{p}_{ij})$;

2. if $C_{ij}^{(s)} = 1$, draw $Y_{ij}^{(s)}$ from the fitted severity distribution;
3. compute ground-up loss $G_{ij}^{(s)} = Y_{ij}^{(s)} V_i$;
4. apply deductible: $L_{ij}^{\text{insured},(s)} = \max(0, G_{ij}^{(s)} - D_i)$.

This is more correct because deductibles are nonlinear and should be applied to realised losses, not only to expected losses.

6 Portfolio aggregation and risk metrics

For hurricane event j , aggregate property-level insured losses:

$$\hat{L}_j^{\text{portfolio}} = \sum_i \hat{L}_{ij}^{\text{insured}}.$$

Across a catalogue of historical or hypothetical events, compute:

$$\text{AAL} = \sum_j q_j \hat{L}_j^{\text{portfolio}},$$

where q_j is the annual event probability or empirical event weight. If no event probabilities are supplied, report unweighted event loss summaries and be explicit that this is not a true stochastic event-set AAL.

For an exceedance probability curve:

$$\text{AEP}(x) = \Pr(L^{\text{portfolio}} > x).$$

Return-period loss is defined by:

$$\Pr(L^{\text{portfolio}} > L_T) = \frac{1}{T}.$$

For example, a 100-year loss is the 99th percentile of the annual loss distribution under the event set.

Coefficient of variation:

$$\text{CV} = \frac{\sigma_L}{\text{AAL}}.$$

This gives a compact measure of portfolio volatility relative to expected loss.

7 Why this model is the right choice

Reason	Explanation
Matches insurance logic	Separates claim occurrence from claim size. This is natural for sparse claims data with many zero losses.
Explainable	Logistic frequency plus Gamma severity can be presented clearly to non-specialists.
Uses cat-model structure	Hazard variables come from storm physics; vulnerability variables come from property characteristics; financial module applies policy terms.
Consistent with Grant & Atkinson	Their main lesson is that building variables alone explain little; wind/location features must be included.
Consistent with Grossi & TeHennepe	Implements event, hazard, vulnerability, financial-loss architecture in a simplified empirical form.
Robust for internship scope	Avoids over-engineered wind-field physics or black-box ML that would be hard to validate and defend.
Production-relevant	Produces property-level scores, event losses, portfolio aggregation, confidence intervals, and sensitivity tests.

8 Validation plan

Validation should be event-aware. Random row splits can leak storm information because the same hurricane appears in many rows.

Recommended validation:

- **Hold out entire hurricanes:** train on some events, test on unseen events.
- **Frequency metrics:** AUC, calibration by probability decile, Brier score.
- **Severity metrics:** MAE/RMSE on positive claims, deviance, calibration of predicted vs actual loss ratio by bins.
- **Portfolio metrics:** predicted vs actual event loss, ranking of high-loss events, error in aggregate portfolio loss.
- **Robustness:** compare with and without hazard features; compare with and without building features; test sensitivity to deductible treatment and outlier caps.

Expected result: building-only models should perform poorly at aggregate level. Adding hazard/distance/track features should materially improve event-level and portfolio-level performance. If it does not, that is an important finding about feature construction or data limitations.

9 Model limitations to state clearly

- Track data are snapshots, not a full physical wind-field simulation.
- Wind at property level is approximated from storm centre/radii variables, not directly observed.
- Claims reflect insurance outcomes, not pure physical damage.
- Deductibles create censoring: some physical losses may not appear as claims.
- Water/storm surge may be mixed into hurricane claims but not explicitly identified.
- If event probabilities are unavailable, AAL/EP curves are scenario-based rather than true vendor-style stochastic outputs.

10 Supervisor-level summary

The proposed model is a hybrid physical-statistical catastrophe model. The physical part constructs hazard features from hurricane track data, mainly distance, wind, pressure, storm size, and quadrant exposure. The statistical part estimates property-level claim probability and conditional loss ratio using interpretable GLMs. The financial part applies replacement value and deductibles, then aggregates losses across the portfolio.

The core equation is:

$$\mathbb{E}[L_{ij}^{\text{ground}}] = \Pr(C_{ij} = 1 \mid H_{ij}, X_i) \times \mathbb{E}\left[\frac{L_{ij}}{V_i} \mid C_{ij} = 1, H_{ij}, X_i\right] \times V_i.$$

This model is simple enough to implement and explain, but rich enough to reflect the main catastrophe modelling logic: hazard drives event severity, building characteristics modify vulnerability, and policy terms convert physical/ground-up loss into insured portfolio loss.